SCHOOL OF COMPUTER SCIENCE & IT, DAVV

CS: 6418 – Advanced Database Management System TEST-2 MTech I (CS/NM/SE)

Duration: 1 Hour Marks 20 Date 31-10-26

1. Consider following relations: *Student*(stud_id, name, dept_name, city, marks), *Course*(c_id,c_name,s_id)
   a. Check marks must be greater than or equal to 0.
   b. Find the total marks obtained by students in each department.
   c. Display all students, including those who are not enrolled in any course.
   d. Find students who live in the same city as Neha.
   e. Find students whose marks are greater than every student in the IT department.
   f. Find the third-highest distinct marks.
   g. Find the name of city where more than 5 students live.
2. Given relation R(A,B,C,D,E,F) with dependencies AB→C,C→D,D→B,BE→F,F→A. Find all candidate keys using attribute closure.
3. Define normalization. Explain insertion, deletion, and update anomalies with examples.